

4

Centre of Gravity and Moment of Inertia

Centre of Gravity of a body

The resultant of the weight of all particles of a body is assumed to be passing through a single point for all positions of the body and that point is called Centre of Gravity (C.G).

Planar Surfaces

A planar surface or a lamina can possess centroid or centre of gravity depending upon whether it is assumed as a massless geometric area or a body with definite mass.

Centre of Gravity of a Lamina

A lamina with definite mass can be assumed to consist of infinite number of particles in the plane of the lamina.

Weights of these individual particles form a system of parallel forces as shown

$$m_1g + m_2g + m_3g + \dots = Mg$$

where Mg is the total weight of the lamina.

Mg is assumed to pass through $G(\bar{x}, \bar{y})$ sum of moments of coplanar forces will be equal to the moment of the resultant.

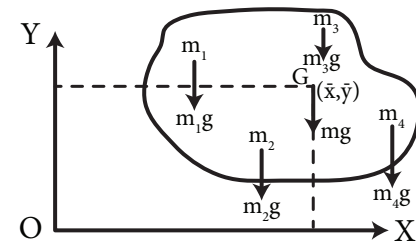
Taking moment about O ,

$$m_1g x_1 + m_2g x_2 + m_3g x_3 + \dots = M \bar{x}$$

$$\bar{x} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + \dots}{M}$$

Imagining the lamina and the reference axis turned by 90° , it can be shown that

$$\bar{y} = \frac{m_1y_1 + m_2y_2 + m_3y_3 + \dots}{M}$$



Centroid of a Uniform Lamina

Uniform lamina has all its particles possessing equal weights. Centroid is the point where the entire area is assumed to be concentrated.

Let the area of the planar surface consists of the number of elemental areas. $a_1, a_2, a_3, \dots$ etc

According to Varignon's theorem, (1) and (2) should be equal.

$$\therefore, a_1x_1 + a_2x_2 + a_3x_3 + \dots = (a_1 + a_2 + a_3 + \dots)\bar{x}$$

$$\text{or, } \bar{x} = \frac{\sum ax}{\sum a} = \frac{\int(ax)}{\int a}$$

By similar reasoning,

$$\bar{y} = \frac{\sum ay}{\sum a} = \frac{\int(ay)}{\int a}$$

Also, centroid of a symmetric lamina falls on its geometric centre.

(1) Centroid of a rectangle

Area of a thin strip of thickness dy at distance y from the x-axis is given by $dA = b \times dy$

$$\begin{aligned} \bar{y} &= \frac{\int y dA}{\int dA} \\ &= \frac{\int_0^d y b dy}{\int_0^d b dy} = \frac{b \left[\frac{y^2}{2} \right]_0^d}{b \left[y \right]_0^d} = \frac{\frac{bd^2}{2}}{bd} = \frac{d}{2} \end{aligned}$$

$$\text{ie, } \bar{y} = \frac{d}{2}$$

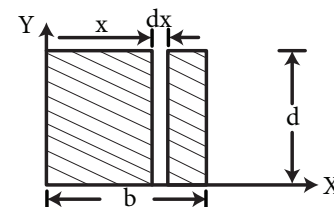
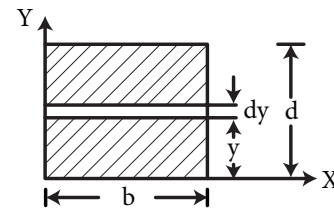
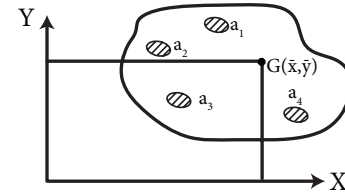
Similarly the elemental area of a vertical strip is given by

$$dA = d \times dx$$

$$\bar{x} = \frac{\int x dA}{\int dA}$$

$$= \frac{\int_0^b x d dx}{\int_0^b d dx} = \frac{b}{2}$$

$$\text{ie, } \bar{x} = \frac{b}{2}$$



The coordinates of the centroid is $\left(\frac{b}{2}, \frac{d}{2}\right)$

(2) *Right angle triangle*

Area of the elemental strip at a distance y from the x -axis is given by

$$dA = b' \times dy$$

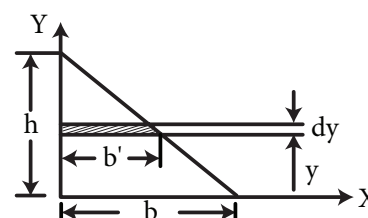
$$\frac{h}{b} = \frac{h-y}{b'} \quad (\text{similar triangles})$$

$$b' = \frac{b(h-y)}{h}$$

$$dA = \frac{b(h-y)}{h} dy$$

$$\bar{y} = \frac{\int y dA}{\int dA} = \frac{\int_0^h y \frac{b(h-y)}{h} dy}{\int_0^h \frac{b(h-y)}{h} dy} = \frac{h}{3}$$

$$\text{ie, } \bar{y} = \frac{h}{3}$$



Similarly, considering a vertical strip and following the same steps, we get

$$\bar{x} = \frac{b}{3}$$

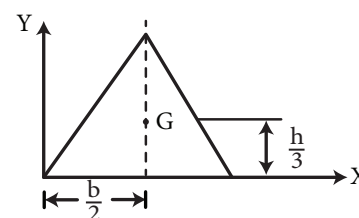
Therefore the coordinates of the centroid are $\left(\frac{b}{3}, \frac{h}{3}\right)$

Isosceles Triangle

Since it has horizontal symmetry,

$$\bar{x} = \frac{b}{2}$$

$$\bar{y} = \frac{h}{3} \text{ can be found using the previous method.}$$

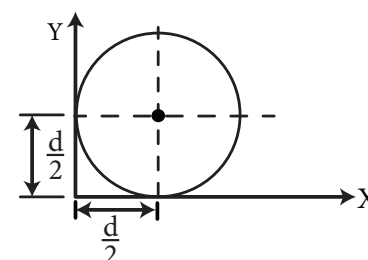


Circle

Both horizontal and vertical symmetry

$$\text{So, } \bar{x} = \frac{d}{2}$$

$$\bar{y} = \frac{d}{2}$$



Semicircle

Since there is horizontal symmetry with respect to y-axis, Centroid is on the y-axis

ie, $\bar{x} = 0$

To find $\bar{y}$, consider an infinitesimal area dA at a radius r as shown

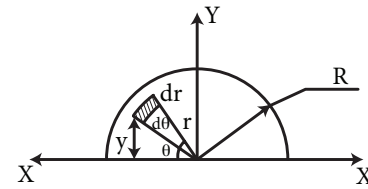
From the figure, $dA = r d\theta dr$ and $y = r \sin \theta$

$$\bar{y} = \frac{\int y dA}{\int dA}$$

$$= \frac{\int_0^\pi \int_0^R r \sin \theta r d\theta dr}{\int_0^\pi \int_0^R r dr d\theta}$$

Integrating and simplifying,

$$\bar{y} = \frac{4R}{3\pi}$$

*Quarter Circle*

Area of the elemental area

$$dA = r d\theta dr$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\bar{x} = \frac{\int x dA}{\int dA}$$

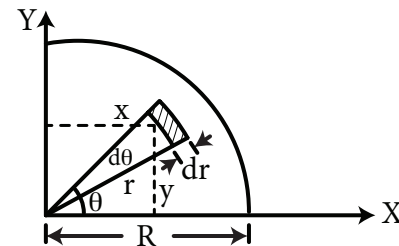
$$= \frac{\int_0^{\pi/2} \int_0^R r \cos \theta r d\theta dr}{\int_0^{\pi/2} \int_0^R r dr d\theta}$$

Integrating and simplifying,

$$\bar{x} = \frac{4R}{3\pi}$$

Similarly,

$$\bar{y} = \frac{4R}{3\pi}$$



Centroid of Composite Areas

The lamina can be split into small definite areas $a_1, a_2, a_3, \dots$ etc.

$$\bar{x} = \frac{\sum ax}{\sum a} \text{ and } \bar{y} = \frac{\sum ay}{\sum a}$$

Problem 1

Locate the centroid of the lamina given below.

All dimensions are in mm .

Solution

Since the lamina has horizontal symmetry, the centroid will fall on a vertical line through the centre, ie, $\bar{x} = 50 mm$

To find $\bar{y}$

$$A_1 = \text{Area of rectangle } ABCD = 100 \times 20 = 2000 mm^2$$

Area of rectangle EFGH

$$A_2 = 60 \times 20 = 1200 mm^2$$

Centroid of $ABCD = C_1$ (10 mm from x-axis)

Centroid of $EFGH = C_2$ ($20 + \frac{60}{2} = 50 mm$ from x-axis)

$$\text{ie, } y_1 = 10 mm$$

$$y_2 = 50 mm$$

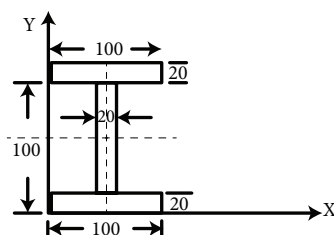
$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A}$$

$$= \frac{(2000)(10) + (1200)(50)}{2000 + 1200} = 25 mm$$

Coordinates of centroid of the entire lamina given by (50, 25)

Problem 2

Locate the centroid of the I-section given below

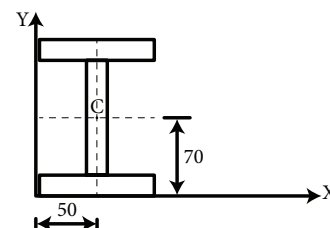
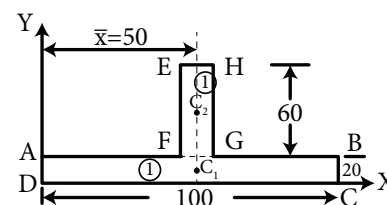
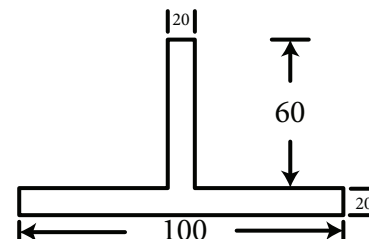


Solution

The section has both vertical and horizontal symmetry

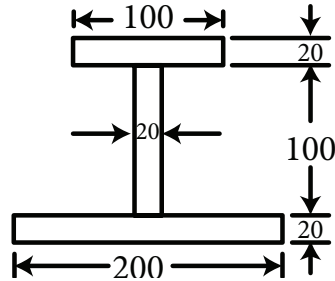
Hence, $\bar{x} = 50$

$$\bar{y} = 70$$



Problem 3

Locate the centroid of the I-section given below
All dimension in mm .

**Solution**

Split the lamina into 3 rectangle ABCD, EFGH and IJKL as shown.

$$\text{Area of } ABCD = A_1 = 200 \times 20 = 4000 \text{ mm}^2$$

$$\text{Area of } EFGH = A_2 = 100 \times 20 = 2000 \text{ mm}^2$$

$$\text{Area of } IJKL = A_3 = 100 \times 20 = 2000 \text{ mm}^2$$

Since there is horizontal symmetry, centroid will fall on the vertical line that divides the section into two symmetrical halves.

Therefore, $\bar{x} = 100 \text{ mm}$ as shown

To find $\bar{y}$

Centroid of ABCD, ie, C_1 is 10 mm above x-axis ie, $y_1 = 10 \text{ mm}$

Centroid of EFGH, ie, C_2 is $20 + \frac{100}{2} = 70 \text{ mm}$ above x-axis

ie, $y_2 = 70 \text{ mm}$

Centroid of IJKL, ie, C_3 is $20 + 100 + \frac{20}{2} = 130 \text{ mm}$ above x-axis

ie, $y_3 = 130 \text{ mm}$

$$\text{Therefore, } \bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3}$$

$$= \frac{(4000)(10) + (2000)(70) + 2000(130)}{4000 + 2000 + 2000} = 55 \text{ mm}$$

Coordinates of the centroid of the entire section are given by

$$(\bar{x}, \bar{y}) = (50, 55)$$

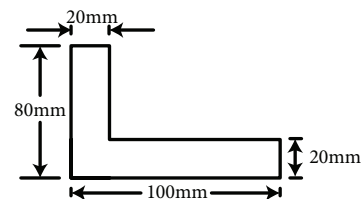
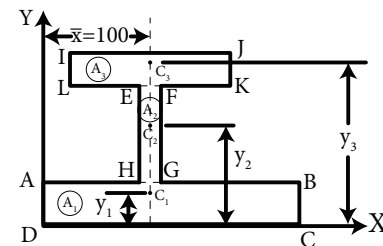
Problem 4

Locate the centroid of the given lamina

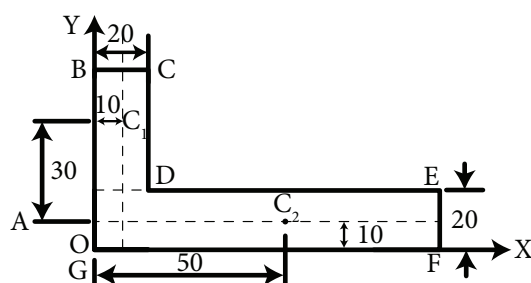
Solution

As there is no symmetry, both $\bar{x}$ and $\bar{y}$ need to be calculated.

$$\text{Area } ABCD = A_1 = 60 \times 20 = 1200 \text{ mm}^2$$



$$\text{Area } A_{EFG} = A_2 = 100 \times 20 = 2000 \text{ mm}^2$$



Coordinates of C₁ and C₂

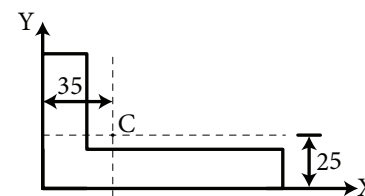
C₁ (10, 50) and C₂ (50, 10) from the figure because C₁ and C₂ will be the geometric centres of the two rectangles ABCD and AEFB.

This means $x_1 = 10, x_2 = 50$

$$y_1 = 50, y_2 = 10$$

$$\begin{aligned} \bar{x}_1 &= \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} \\ &= \frac{(1200)(10) + (2000)(50)}{1200 + 2000} = 35 \text{ mm} \end{aligned}$$

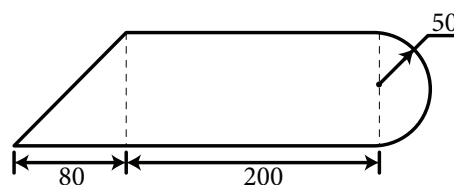
$$\begin{aligned} \bar{y}_1 &= \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} \\ &= \frac{(1200)(50) + 2000(10)}{1200 + 2000} = 25 \text{ mm} \end{aligned}$$



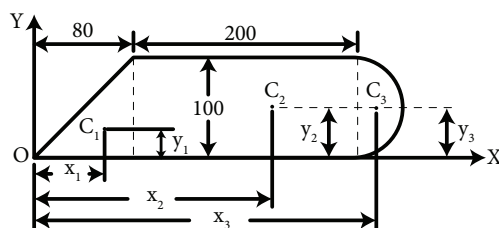
So, the coordinates of the centroid are given by (35, 25)

Problem 5

Locate the centroid of the composite area given below
All dimensions in mm.



Solution



$$\text{Area of triangle } A_1 = \frac{1}{2}(80)(100) = 4000 \text{ mm}^2$$

$$\text{Area of rectangle } A_2 = 200 \times 100 = 20000 \text{ mm}^2$$

$$\text{Area of semicircle } A_3 = \frac{1}{2} \times \frac{\pi}{4} (100)^2 = 3925 \text{ mm}^2$$

From the geometry, the centroids of the three areas can be located as follows

Centroid of triangle, (C_1)

$$x_1 = 80 - \frac{80}{3} = 53.33$$

$$y_1 = \frac{100}{3} = 33.33$$

Centroid of rectangle (C_2)

$$x_2 = 80 + \frac{200}{2} = 180 \text{ mm}$$

$$y_2 = \frac{100}{2} = 50 \text{ mm}$$

Centroid of semicircle (C_3)

$$x_3 = 80 + 200 + \frac{4(50)}{3\pi} = 301.23 \text{ mm}$$

$$y_3 = \frac{100}{2} = 50 \text{ mm}$$

Now,

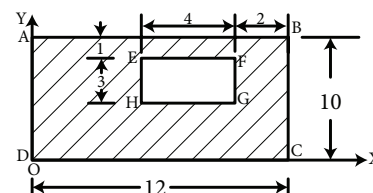
$$\begin{aligned} \bar{x} &= \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3} \\ &= \frac{4000(53.33) + 20000(180) + 3925(301.23)}{4000 + 20000 + 3925} \\ &= 179 \text{ mm} \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} \\ &= \frac{4000(33.33) + 20000(50) + 3925(50)}{4000 + 2000 + 3925} \\ &= 47.6 \text{ mm} \end{aligned}$$

So, the location of the centroid of the composite area is given by the coordinates (179, 47.6)

Problem 6

A rectangular hole EFGH has been cut in a lamina ABCD as shown. Find the centroid of the remaining area. (All dimensions in *cm*).

**Solution**

In the expression $\sum A_y$, $\sum A_x$ & $\sum A$, the term containing area EFGH has to be subtracted.

$$\text{Area ABCD, } A_1 = 12 \times 10 = 120 \text{ cm}^2$$

$$\text{Area EFGH, } A_2 = 4 \times 3 = 12 \text{ cm}^2$$

Centroid of ABCD, C_1 is given by (6, 5)

ie, $x_1 = 6$ and $y_1 = 5$

Centroid of EFGH is given by

$$x_2 = 6 + \frac{4}{2} = 8 \text{ cm and } y_2 = 6 + \frac{3}{2} = 7.5 \text{ cm}$$

ie, (8, 7.5)

Coordinates of the centroid of the set (remaining) area can be found as

$$\begin{aligned} \bar{x} &= \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2} \\ &= \frac{(120)(6) - (12)(8)}{120 - 12} = 5.77 \text{ cm} \\ \bar{y} &= \frac{A_1 y_1 - A_2 y_2}{A_1 - A_2} \\ &= \frac{(120)(5) - (12)(7.5)}{120 - 12} = 4.72 \text{ cm} \\ \therefore, \text{ Location of } C & (5.77, 4.72) \end{aligned}$$

Problem 7

Find the location of the centroid of the given section. All dimensions are in *cm*.

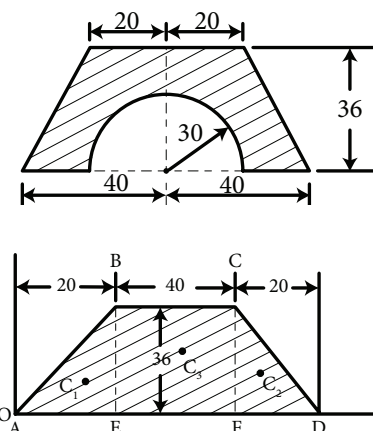
Solution

Step 1 Locate the centroid of the full trapezium ABCD

$$A_1 = \text{Area of triangle } ABE = \frac{1}{2} (20)(36) = 360 \text{ cm}^2$$

$$A_2 = \text{Area of triangle } FCD = \frac{1}{2} (20)(36) = 360 \text{ cm}^2$$

$$A_3 = \text{Area of rectangle } BCFE = 36 \times 40 = 1440 \text{ cm}^2$$



Centroid of triangle 1 (C_1)

$$x_1 = 20 - \frac{20}{3} = 13.33 \text{ cm}$$

$$y_1 = \frac{36}{3} = 12 \text{ cm}$$

Centroid of triangle 2 (C_2)

$$x_2 = 20 + 40 + \frac{20}{3} = 66.67 \text{ cm}$$

$$y_2 = \frac{36}{3} = 12 \text{ cm}$$

Centroid of rectangle (C_3)

$$x_3 = 20 + \frac{40}{2} = 40 \text{ cm}$$

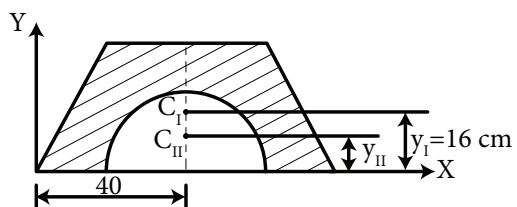
$$y_3 = \frac{36}{2} = 18 \text{ cm}$$

Centroid(C) of the entire trapezium ABCD is given by

$$\begin{aligned} \bar{x} &= \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3} \\ &= \frac{(360)(13.33) + (360)(66.67) + (1440)(40)}{360 + 360 + 1440} \\ &= 40 \text{ cm} \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} \\ &= \frac{(360)(12) + (360)(12) + (1440)(18)}{360 + 360 + 1440} \\ &= 16 \text{ cm} \end{aligned}$$

Centroid of the trapezium with semi-circular hole



Since the lamina has horizontal symmetry we have

$$\bar{x} = 40 \text{ cm}$$

To find $\bar{y}$

Area of trapezium, $A_I = 360 + 360 + 1440 = 2160 \text{ cm}^2$

Area of hole, $A_{II} = \frac{1}{2} \times \frac{\pi}{4} (60)^2 = 1413 \text{ cm}^2$

Centroidal distance of trapezium, $y_I = 16 \text{ cm}$

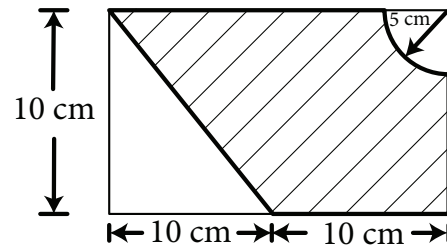
Centroidal distance of semicircular hole, $y_{II} = \frac{4(30)}{3\pi}$
 $= 12.73 \text{ cm}$

$$\begin{aligned} \therefore \bar{y} &= \frac{A_I y_I - A_{II} y_{II}}{A_I - A_{II}} \\ &= \frac{(2160)(16) - (1413)(12.73)}{2160 - 1413} \\ &= 22.2 \text{ cm} \end{aligned}$$

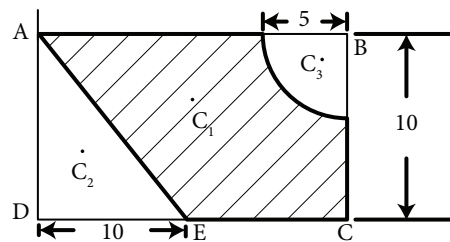
Coordinates of the section is given by C (40, 22.2)

Problem 8

Find the centroid of the shaded area given in the figure below



Solution



Centroid of the rectangle ABCD (C_1)

$$x_I = 10 \text{ cm}$$

$$y_I = 5 \text{ cm}$$

Centroid of the triangle ADE (C_2)

$$x_2 = \frac{10}{3} = 3.33 \text{ cm}$$

$$y_2 = \frac{10}{3} = 3.33 \text{ cm}$$

Centroid of the quarter circle (C_3)

$$x_3 = 20 - \frac{4(5)}{3\pi} = 17.9 \text{ cm}$$

$$y_3 = 10 - \frac{4(5)}{3\pi} = 7.88 \text{ cm}$$

Area of ABCD, $A_1 = 20 \times 10 = 200 \text{ cm}^2$

Area of ADE, $A_2 = \frac{1}{2} \times 10 \times 10 = 50 \text{ cm}^2$

Area of semicircle, $A_3 = \frac{1}{4} \times \frac{\pi}{4} (10)^2 = 19.6 \text{ cm}^2$

The centroid of the shaded portion after areas A_2 and A_3 have been cut off from the area A_1 , can be found as follows

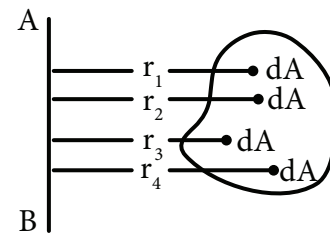
$$\begin{aligned} \bar{x} &= \frac{A_1 x_1 - (A_2 x_2 + A_3 x_3)}{A_1 - (A_2 + A_3)} \\ &= \frac{200(10) - (50 \times 3.33 - 19.6 \times 17.90)}{200 - (50 + 19.6)} = 16.75 \text{ cm} \\ \bar{y} &= \frac{A_1 y_1 - (A_2 y_2 + A_3 y_3)}{A_1 - (A_2 + A_3)} \\ &= \frac{200(5) - (50 \times 3.33 - 19.6 \times 7.88)}{200 - (50 + 19.6)} \\ &= 7.58 \text{ cm} \end{aligned}$$

Coordinates are (16.75, 7.58)

Second Moment of Area

It is also called Area Moment of Inertia. This gives an estimate of how an area is distributed with respect to a given axis.

If an area is split up into infinitesimal areas dA each and if these elemental areas are at distances $r_1, r_2, r_3 \dots$ etc from a given axis AB,



then the sum of all terms $r_1^2 dA + r_2^2 dA + r_3^2 dA + \dots$ is called the second moment of area or Area moment of Inertia.

Moment of inertia of the lamina about the axis AB is given by

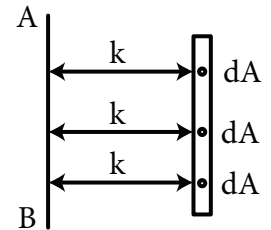
$$I_{AB} = \sum r^2 dA$$

$$= \int r^2 dA$$

If the elemental areas are rearranged to be at equal distance from the axis keeping the moment of inertia I_{AB} constant, then

$$I_{AB} = dAk^2 + dAk^2 + \dots$$

$$= k^2 \sum dA = Ak^2$$



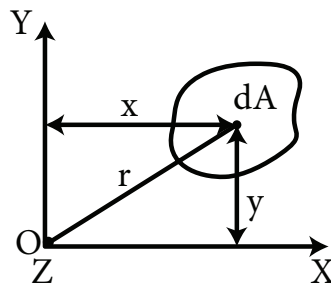
Where k is called Radius of Gyration

So the radius of gyration of a given lamina about any axis is the constant distance from that axis at which the elemental areas of the lamina need to be, without changing the moment of inertia about that axis.

Perpendicular Axis Theorem

If I_{xx} and I_{yy} are the moments of inertia of an area about two mutually perpendicular axis OX and OY in the plane of the area and I_{zz} be the moment of inertia of the area about an axis OZ perpendicular to OX and OY as well as passing through the intersection of OX and OY, then $I_{zz} = I_{xx} + I_{yy}$

Proof



Area A is in the XY plane and z-axis is perpendicular to it. Consider an elemental area dA which is at distances x , y and r from y-axis, x-axis and z-axis respectively.

From the figure, $r^2 = x^2 + y^2$

$$I_{zz} = \int r^2 dA = \int (x^2 + y^2) dA$$

$$= \int x^2 dA + \int y^2 dA$$

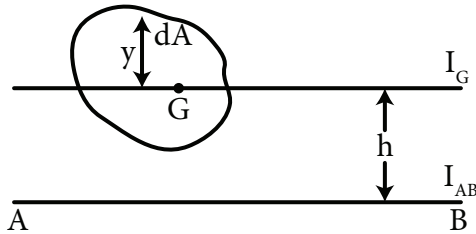
$$= I_{YY} + I_{XX}$$

$$I_{ZZ} = I_{XX} + I_{YY}$$

Parallel Axis Theorem

The moment of inertia of an area about any axis in the same plane will be the sum of its moment of inertia about a parallel centroidal axis in the same plane and the product of the area with the square of the distance between the two axis.

Proof



G is the centroid of the area. Let the distance between the centroidal axis and axis AB be h as shown. Consider an elemental area dA at y distance from the centroidal axis.

$$\begin{aligned} I_{AB} &= \int dA (y + h)^2 = \int y^2 dA + \int 2hy dA + \int h^2 dA \\ &= \int y^2 dA + 2h \int y dA + h^2 \int dA \\ &= I_G + 2h A \bar{y} + h^2 A \end{aligned}$$

But $2h A \bar{y} = 0$ because $\bar{y}$ is zero, since y is measured from the centroidal axis itself.

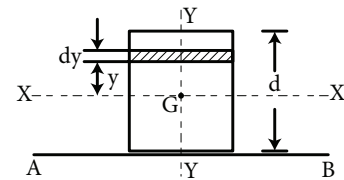
So we have $I_A = I_G + Ah^2$

(1) Moment of Inertia of a rectangular area

About centroidal axis

Consider an elemental strip at y distance from the centroidal axis XX

$$\begin{aligned} I_{GXX} &= \int_{-d/2}^{+d/2} y^2 dA = \int_{-d/2}^{+d/2} y^2 b dy \\ &= \left[\frac{y^3}{3} \right]_{-d/2}^{+d/2} b dy = \frac{bd^3}{12} \\ I_{GXX} &= \frac{bd^3}{12} \end{aligned}$$



Similarly it can be shown that the moment of inertia about the centroidal axis Y-Y

$$= I_{GY} = \frac{db^3}{12}$$

About the base

Referring to the figure...., the moment of inertia about the axis AB along the base of the rectangle can be found out using parallel axis theorem.

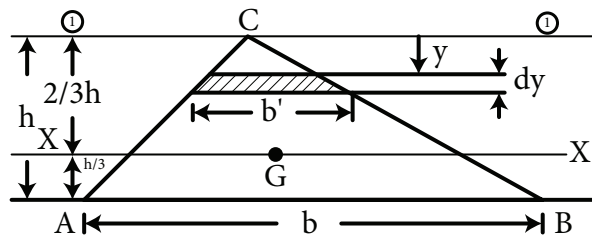
$$I_{AB} = I_{GXX} + Ah^2$$

Where h is the distance between the axis XX and AB which is $\frac{d}{2}$

$$\begin{aligned} \text{So, } I_{AB} &= I_{GXX} + A\left(\frac{d}{2}\right)^2 \\ &= \frac{bd^3}{12} + (bd)\left(\frac{d}{2}\right)^2 \\ &= \frac{bd^3}{3} \end{aligned}$$

$$I_{AB} = \frac{bd^3}{3}$$

(2) Moment of Inertia of triangular area



About axis (1)-(1)

Let the elemental strip dA be at a distance y from the axis (1)-(1) passing through the top vertex C.

$$\begin{aligned} I_{1-1} &= \int_0^h y^2 dA \\ &= \int_0^h y^2 b' dy \end{aligned}$$

But from similar triangles,

$$\frac{b'}{y} = \frac{b}{h} \text{ or } b' = \frac{by}{h}$$

Substituting,

$$I_{l-l} = \int_0^h \frac{b}{h} y^3 dy = \frac{bh^3}{4}$$

$$I_{l-l} = \frac{bh^3}{4}$$

About centroidal axis X-X

Consider the two parallel axis $l-l$ and $X-X$ which apart by a distance $\frac{2}{3}h$.

Involving parallel axis theorem,

$$I_{l-l} = I_{GXX} + A \left(\frac{2}{3}h \right)^2$$

$$\begin{aligned} \text{or, } I_{GXX} &= I_{l-l} - A \left(\frac{2}{3}h \right)^2 \\ &= \frac{bh^3}{4} - \left(\frac{1}{2}bh \right) \left(\frac{2}{3}h \right)^2 = \frac{bh^3}{36} \\ I_{GXX} &= \frac{bh^3}{36} \end{aligned}$$

About the centroidal axis perpendicular to the plane.

Let Z-Z be an axis normal to the area and passing through G.

Then I_{GZZ} can be found using perpendicular axis theorem.

$$I_{ZZ} = I_{XX} + I_{YY} = \frac{\pi}{64} D^4 + \frac{\pi}{64} D^4 = \frac{\pi}{32} D^4$$

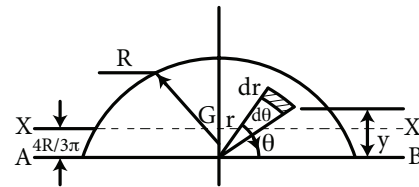
$$\text{ie, } I_{ZZ} = \frac{\pi}{32} D^4$$

This is known as polar moment of Inertia

(4) Semi-circular Area

About base AB

$$\begin{aligned} I_{AB} &= \int y^2 dA \\ &= \int_0^\pi \int_0^R (r \sin \theta)^2 r dr d\theta = \frac{\pi R^4}{8} = \frac{\pi D^4}{128} \\ I_{AB} &= \frac{\pi D^4}{128} \end{aligned}$$



About centroidal axis X-X

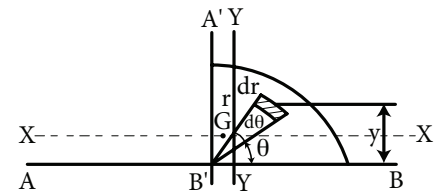
Using parallel axis theorem,

$$\begin{aligned}
 I_{GXX} &= I_{AB} - Ah^2 \\
 &= \frac{\pi D^4}{128} - \frac{1}{2} \left(\frac{\pi D^2}{4} \right) \left(\frac{4R}{3\pi} \right)^2 \\
 &= R^4 \left[\frac{\pi}{8} - \frac{8}{9\pi} \right] = 0.11R^4 \\
 I_{GXX} &= 0.11R^4
 \end{aligned}$$

(5) Quarter-circle

About bases

$$\begin{aligned}
 I_{AB} &= \int y^2 dA \\
 &= \int_0^{\pi/2} \int_0^R (r \sin \theta)^2 r d\theta dr \\
 &= \frac{\pi R^4}{16} = \frac{\pi D^4}{256}
 \end{aligned}$$



Similarly, $I_{A'B'} = \frac{\pi D^4}{256}$

ie, $I_{AB} = I_{A'B'} = \frac{\pi D^4}{256}$

About centroidal axis

Using parallel axis theorem,

$$\begin{aligned}
 I_{GXX} &= I_{AB} - Ah^2 \\
 &= \frac{\pi D^4}{256} - \frac{1}{4} \left(\frac{\pi D^2}{4} \right) \left(\frac{4R}{3\pi} \right)^2 \\
 &= R^4 \left[\frac{\pi}{16} - \frac{4}{9\pi} \right] = 0.055R^4
 \end{aligned}$$

Similarly, $I_{GYY} = 0.055R^4$

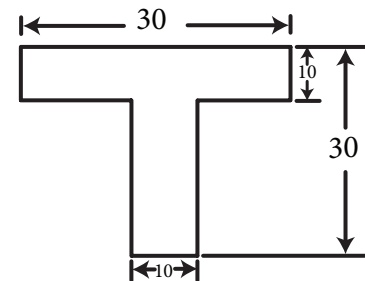
ie, $I_{GXX} = I_{GYY} = 0.055R^4$

Problems 9

Find out the moment of inertia of the given T-section about its vertical centroidal axis. All dimension is *cm*.

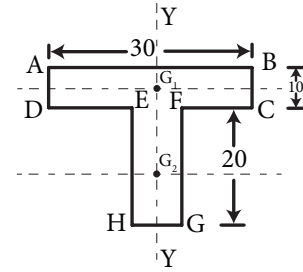
Solution

Divide the T-section into two rectangle ABCD and EFGH. ABCD, EFGH and the T-section itself are symmetric with respect to Y-Y axis. Hence the centroids G_1 and G_2 as well as the centroid of the T-section will fall on the Y-Y axis.



Therefore I_{GYG} of the T-section can be found out simply by adding I_{G1YY} and I_{G2YY}

$$\begin{aligned}
 I_{GYG} &= I_{G1YY} + I_{G2YY} \\
 &= \frac{d_1 b_1^3}{12} + \frac{d_2 b_2^3}{12} \\
 &= \frac{(10)(30)^3}{12} + \frac{(20)(30)^3}{12} \\
 &= 22500 + 1666.7 = 24166.7 \text{ cm}^4
 \end{aligned}$$

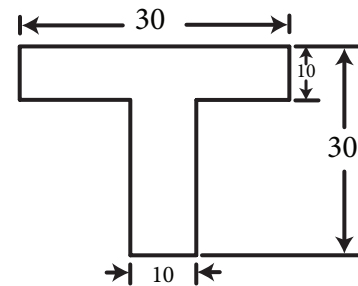
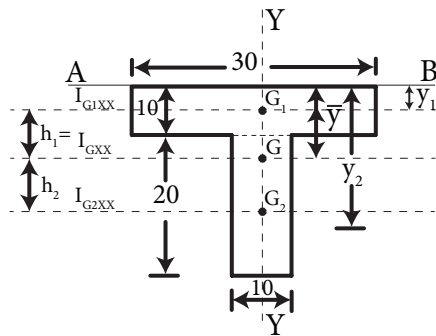


Problem 10

Find out the moment of inertia of the given T-section about its horizontal centroidal axis.

All dimensions in *cm*.

Solution



Here the centroidal axis of ABCD and EFGH are two parallel horizontal lines. So we cannot simply add the individual moments of inertia.

We have to use parallel axis theorem to transfer the M.Is to the centroidal axis of the whole T-section and then add them.

To find the centroid (G) of T-section

Let the distance of G from the top (A-B) be $\bar{y}$

$$\text{From the figure, } y_1 = \frac{10}{2} = 5 \text{ cm}$$

$$y_2 = 10 + \frac{20}{2} = 20 \text{ cm}$$

$$A_1 = 10 \times 30 = 300 \text{ cm}^2$$

$$A_2 = 10 \times 30 = 200 \text{ cm}^2$$

$$\text{Then, } \bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

$$= \frac{(300)(5) + (200)(20)}{300 + 200}$$

$$= 11 \text{ cm}$$

Locate G at 11cm from the top (A-B)

Referring to the figure, h_1 and h_2 are the distances of the individual centroidal axis from the centroidal axis of the T-section. Transfer the individual M. Is, I_{G1XX} and I_{G2XX} to the centroidal axis of the T-section using parallel axis theorem and then add them.

$$h_1 = \bar{y} - y_1 \text{ and } h_2 = y_2 - \bar{y}$$

$$I_{GXX} = (I_{G1XX} + A_1 h_1^2) + (I_{G2XX} + A_2 h_2^2) \quad (1)$$

$$I_{G1XX} = \frac{b_1 d_1^3}{12} = \frac{30 \times 10^3}{12} = 2500 \text{ cm}^4$$

$$I_{G2XX} = \frac{b_2 d_2^3}{12} = \frac{(10)(20)^3}{12} = 6666.7 \text{ cm}^4$$

$$A_1 = 30 \times 10 = 300 \text{ cm}^2; A_2 = 20 \times 10 = 200 \text{ cm}^2$$

$$h_1 = \bar{y} - y_1 = 11 - 5 = 6 \text{ cm}$$

$$h_2 = y_2 - \bar{y} = 20 - 11 = 9 \text{ cm}$$

Substituting in (1),

$$I_{GXX} = [(2500) + (300 \times 6^2)] + [(6666.7) + (200 \times 9^2)]$$

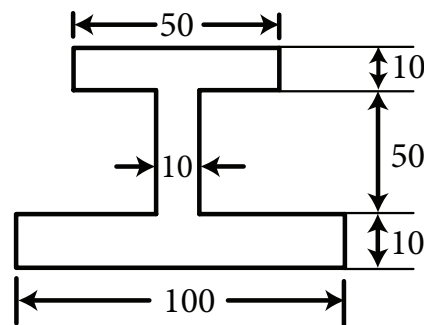
$$= 13300 + 22866.7$$

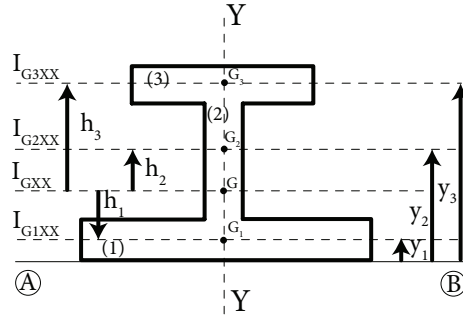
$$= 36166.7 \text{ cm}^4$$

Problem 11

Determine the moment of inertia of the given I-section about its two centroidal axis.

All dimensions are in mm.



Solution

For rectangle (1), $I_{G1XX} = \frac{b_1 d_1^3}{12} = \frac{(100)(10)^3}{12} = 8333.3 \text{ mm}^4$

For rectangle (2), $I_{G2XX} = \frac{b_2 d_2^3}{12} = \frac{(10)(50)^3}{12} = 104166.7 \text{ mm}^4$

For rectangle (3), $I_{G3XX} = \frac{b_3 d_3^3}{12} = \frac{(50)(10)^3}{12} = 4166.7 \text{ mm}^4$

To find I_{GXX}

The distances of centroids G_1 , G_2 and G_3 of the three rectangle from the base (axis $A - B$) are found from the figure

$$y_1 = \frac{10}{2} = 5 \text{ mm}$$

$$y_2 = 10 + \frac{50}{2} = 35 \text{ mm}$$

$$y_3 = 10 + 50 + \frac{10}{2} = 65 \text{ mm}$$

Centroidal distance of the entire T-section from the base can be found as follows

$$\begin{aligned} \bar{y} &= \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} \\ &= \frac{(100 \times 10)(5) + (50 \times 10)(35) + (50 \times 10)(65)}{(100 \times 10) + (50 \times 10) + (50 \times 10)} \\ &= 27.5 \text{ mm} \end{aligned}$$

Locate G at 27.5 mm from the base

From the geometry,

$$h_1 = \bar{y} - y_1 = 27.5 - 5 = 22.5 \text{ mm}$$

$$h_2 = y_2 - \bar{y} = 35 - 27.5 = 7.5 \text{ mm}$$

$$h_3 = y_3 - \bar{y} = 65 - 27.5 = 37.5 \text{ mm}$$

Transferring I_{G1XX} , I_{G2XX} and I_{G3XX} to the centroidal axis through G using parallel axis theorem and then adding them,

$$\begin{aligned} I_{GXX} &= [I_{G1XX} + A_1 h_1^2] + [I_{G2XX} + A_2 h_2^2] + [I_{G3XX} + A_3 h_3^2] \\ &= [8333.3 + (100)(10)(22.5)^2] + [104166.7 + (50)(10)(7.5)^2] \\ &\quad + [4166.7 + (50)(10)(37.5)^2] \\ &= 514583.3 + 132291.7 + 707291.7 \\ &= 1354166.7 \text{ mm}^4 \end{aligned}$$

To find I_{GYY}

$$\text{For rectangle (1), } I_{G1YY} = \frac{d_1 b_1^3}{12} = \frac{(10)(100)^3}{12} = 833333.3 \text{ mm}^4$$

$$\text{For rectangle (2), } I_{G2YY} = \frac{d_2 b_2^3}{12} = \frac{(50)(10)^3}{12} = 4166.7 \text{ mm}^4$$

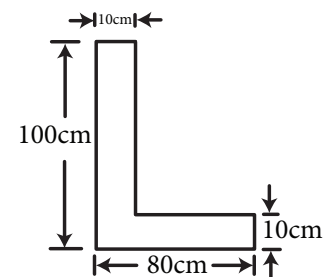
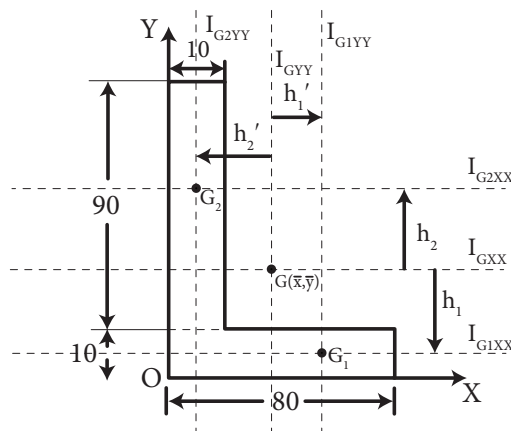
$$\text{For rectangle (3), } I_{G3YY} = \frac{d_3 b_3^3}{12} = \frac{(10)(50)^3}{12} = 104166.7 \text{ mm}^4$$

Since G_1 , G_2 , G_3 and G fall on the same centroidal $Y - Y$, axis,

$$\begin{aligned} I_{GYY} &= I_{G1YY} + I_{G2YY} + I_{G3YY} \\ &= 833333.3 + 4166.7 + 104166.7 \\ &= 941666.7 \text{ mm}^4 \end{aligned}$$

Problem 12

Determine the moment of inertia of the angle section given in the figure about the centroidal axis.



To find the centroid $G(\bar{x}, \bar{y})$ of the L-section

$$x_1 = \frac{80}{2} 40 \text{ cm} \quad ; \quad x_2 = \frac{10}{2} = 5 \text{ cm}$$

$$y_1 = \frac{10}{2} = 5 \text{ cm} \quad ; \quad y_2 = 10 + \frac{90}{2} = 55 \text{ cm}$$

$$A_1 = 80 \times 10 = 800 \text{ cm}^2 \quad ; \quad A_2 = 90 \times 10 = 900 \text{ cm}^2$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = \frac{(800 \times 40) + (900 \times 5)}{800 + 900} = 21.5 \text{ cm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{(800 \times 5) + (900 \times 55)}{800 + 900} = 31.5 \text{ cm}$$

Locate G (21.5, 31.5)

To find the individual M.Is if rectangle 1 and 2

$$I_{G1XX} = \frac{b_1 d_1^3}{12} = \frac{80(10)^3}{12} = 6666.7 \text{ cm}^4$$

$$I_{G1YY} = \frac{d_1 b_1^3}{12} = \frac{10(80)^3}{12} = 426666.7 \text{ cm}^4$$

$$I_{G2XX} = \frac{d_2 b_2^3}{12} = \frac{10(90)^3}{12} = 607500 \text{ cm}^4$$

$$I_{G2YY} = \frac{b_2 d_2^3}{12} = \frac{90(10)^3}{12} = 7500 \text{ cm}^4$$

To find I_{GXX}

From the figure,

$$h_1 = \bar{y} - y_1 = 31.5 - 5 = 26.5 \text{ cm}$$

$$h_2 = y_2 - \bar{y} = 55 - 31.5 = 23.5 \text{ cm}$$

$$\begin{aligned} I_{GXX} &= (I_{G1XX} + A_1 h_1^2) + (I_{G2XX} + A_2 h_2^2) \\ &= [6666.7 + 800(26.5)^2] + [607500 + 900(23.5)^2] \\ &= 568466.7 + 1104525 = 1672991.7 \text{ cm}^4 \end{aligned}$$

To find I_{GYY}

From the figure,

$$h'_1 = x_1 - \bar{x} = 40 - 21.5 = 18.5 \text{ cm}$$

$$h'_2 = \bar{x} - x_2 = 21.5 - 5 = 16.5 \text{ cm}$$

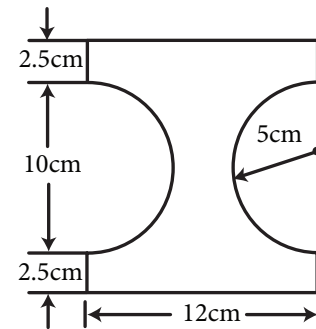
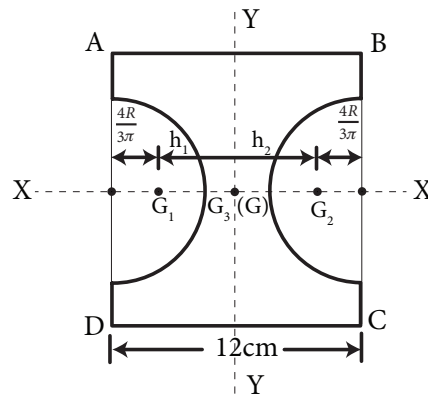
$$\begin{aligned} I_{GYG} &= (I_{G1YY} + A_1 h'_1{}^2) + (I_{G2YY} + A_2 h'_2{}^2) \\ &= [426666.7 + 800(18.5)^2] + [7500 + 900(16.5)^2] \\ &= 700466.7 + 252525 = 952991.7 \text{ cm}^4 \end{aligned}$$

Problem 13

Find out the moment of inertia of the section given about the horizontal and vertical axis passing through the centroid.

Solution

As the section has both vertical and horizontal symmetry, the centroid of the whole section (G) will fall on the geometric centre as shown.



To find I_{GXX}

Centroids of the semi-circular areas (G_1 and G_2) and the centroid of the rectangular area ABCD (G_3)

As well as the centroid (G) of the section fall on $X-X$ axis.

$$\text{Therefore } I_{GXX} = I_{G3XX} - (I_{G1XX} + I_{G2XX})$$

Because the two semicircular areas have to be removed from the rectangular area to get the given section.

$$I_{G3XX} = \frac{bd^3}{12} = \frac{(12)(15)^3}{12} = 3375 \text{ cm}^4$$

$$I_{G1XX} = I_{G2XX} = \frac{\pi D^4}{128} = \frac{\pi (10)^4}{128} = 245.3 \text{ cm}^4$$

Substituting,

$$I_{GXX} = 3375 - (245.3 + 245.3) = 2884 \text{ cm}^4$$

To find I_{GYG}

$$I_{G1YY} = I_{G2YY} = 0.11R^4 = 0.11(5)^4 = 68.75 \text{ cm}^4$$

$$I_{G3YY} = \frac{db^3}{12} = \frac{15(12)^3}{12} = 2160 \text{ cm}^4$$

I_{G1YY} and I_{G2YY} need to be transferred to Y-Y axis using parallel axis theorem.

From the figure, $h_1 = h_2 = 6 - \frac{4R}{3\pi} = 3.88 \text{ cm}$

$$A_1 = A_2 = \frac{1}{2} \left(\frac{\pi}{4} D^2 \right) = \frac{1}{2} \times \frac{\pi}{4} \times 10^2 = 39.25 \text{ cm}^2$$

$$\begin{aligned} I_{GY} &= I_{G3YY} - [I_{G1YY} + A_1 h_1^2] - [I_{G2YY} + A_2 h_2^2] \\ &= 2160 - [68.75 + 39.25(3.88)^2] - [68.75 + 39.25(3.88)^2] \\ &= 840.8 \text{ cm}^4 \end{aligned}$$